Bases and Caps General Notes

a. Uplift and lateral loads have been increased for earthquake or wind loading with no further increase allowed. Reduce where other loads govern.

b. Downloads are based on full bearing of the post on the post base and post cap. Downloads may not be increased for short-term loading, and shall be reduced where limited by the post allowable load. See **strongtie.com/posts** for common post allowable loads.

c. For post bases with 1" standoffs, full bearing on concrete is required. Standoffs shall bear on top of concrete and not be embedded into wet concrete. A higher download may be achieved by solidly packing grout in the 1" standoff area before installation of the post. Allowable download shall be based on the capacity of the post, grout or concrete according to the code. (Figure 1)

d. The designer is responsible for concrete design.

e. For post-installed bases, the designer must specify anchor bolt type, length and embedment. Titen HD®, Stainless-Steel Titen HD and SET-3G® are some of the retrofit anchor options. For more information, refer to our Anchoring, Fastening, Restoration and Strengthening Systems *for Concrete and Masonry* catalog at **strongtie.com**.

f. Except for the MPBZ® moment post base, post bases do not provide adequate resistance to prevent members from rotating about the base and are therefore not recommended for non-top-supported installations (such as fences, patio covers, pergolas or unbraced carports). The top of the post must be restrained from moving horizontally by some other means, e.g., by tying the roof into a supporting structure or by adding knee bracing between the posts and beams. (Figure 2) Alternatively, see the MPBZ on pp. 88–89 for a post base that provides moment resistance for columns or posts. (Figure 3)

g. For post bases that do not attach to all four sides of the post, the post may be wider than the base and overhang the base in one direction (e.g., a 6x8 post on an ABU66Z) as long as the bearing area provided by the base is sufficient for the post.

h. For applications involving the use of a post base to support a wood beam, refer to p. 76. (Figure 4)

i. Structural composite lumber columns have sides that show either the wide face or the edges of the lumber strands/veneers. Allowable loads for caps and bases reflect installation into the wide face. See technical bulletin T-C-SCLCLM at **strongtie.com** for load reductions due to narrow face installations. Some products require installation of fasteners into the wide face only.

j. Unless otherwise noted, allowable downloads for post caps are based on the assumption that the wood post is at least as wide as the supported beam. For applications where this condition is not met (e.g., a 6x10 beam supported by 4x4 post), the designer must evaluate the download capacity.

k. Allowable lateral loads for post caps can only be achieved if one of the members, the post or beam, is supported laterally by other means. (Figure 2) For applications involving lateral load transfer from the beam to the column, the column must be designed to receive the load without rotating about its base (e.g., cantilevered out of the ground, diagonally braced, or connected at the base with a moment-resisting connector like the MPBZ post base).

l. Post cap allowable loads are for a continuous beam. Unless otherwise noted, the beam may be spliced at the centerline of the post, and the maximum allowable download for each spliced beam is one half of the cap's tabulated allowable download. For CC, CCQ™ and CCOS caps, the download for each spliced beam shall also not exceed 2x the download of the other spliced beam. Eccentricity induced in the post by spliced beams must be evaluated by the designer. Tabulated uplift and lateral loads do not apply to spliced beam conditions. When spliced beams must be connected together to transfer design tension loads (lateral loads parallel to the beams), the connection must be by means other than the post cap.

m. Post caps may be installed inverted for post-to-beam applications. The designer is responsible for evaluating the beam to ensure it is capable of receiving all loads applied by the post. (Figure 5)

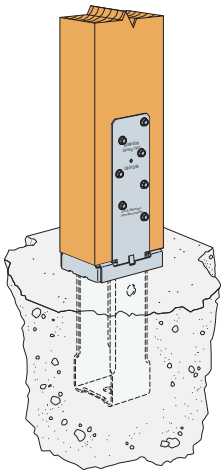


Figure 1 — Post Base with 1" Standoff

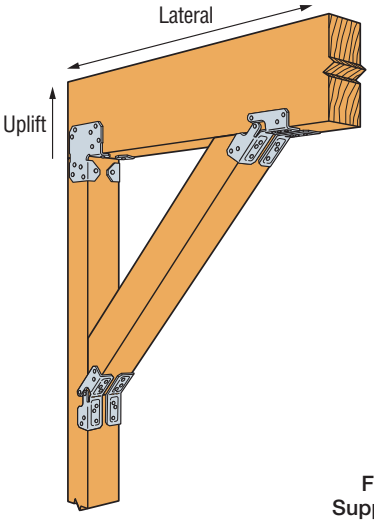


Figure 2 — Typical LCE™ Post Cap Installation with KBS1Z® Knee Bracing

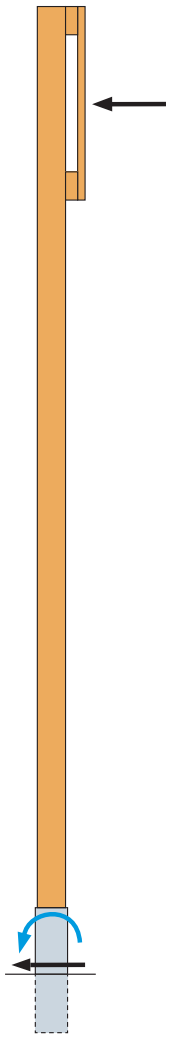


Figure 3 — Non-Top-Supported Post with MPBZ Moment Post Base

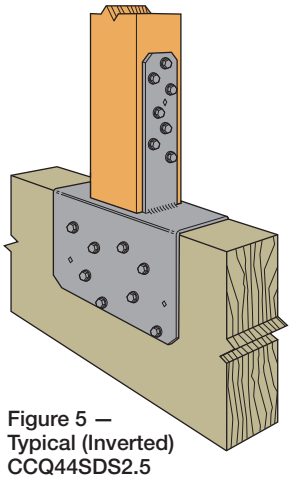


Figure 5 — Typical (Inverted) CCQ44SDS2.5 Post-to-Beam Installation

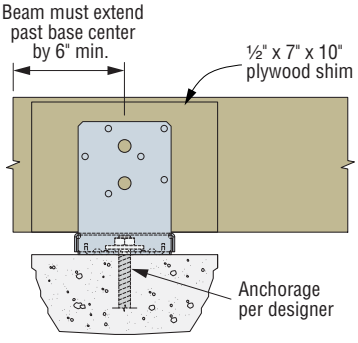


Figure 4 — ABU with Beam Installation



Visit **app.strongtie.com/pbs** to access our Post-to-Beam Selector web application.

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